Isomorphic Questions

Multiple choice questions generated below. Answers in RED and incorrect answers in BLUE.

Q1. Which of the following statements about modeling/approximations is most appropriate?

A. We can treat a human hand as a point particle when analyzing the detailed bending of a finger joint.
This is incorrect because the finger's rotation and deformation require an extended-body model, not a point particle.

B. We can treat a car as a point mass when tracking its change in position over several kilometers on a highway.
This is correct because over kilometer-scale translation the car's size is negligible compared with the distance traveled.

C. We are not allowed to assume a vacuum in any physics problem, even when air drag is intentionally neglected.
This is incorrect because assuming a vacuum is a standard idealization used precisely to ignore air resistance.

D. None of the above statements can ever be correct because idealizations are never justified in physics.
This is incorrect because idealized models are often justified when they capture the dominant physics at the scale of interest.

Q2. Which of the following unit-conversion equations is correct?

A. 1 mm = 0.01 m
This is incorrect because a millimeter is one-thousandth of a meter, not one-hundredth.

B. 1 = 1000
This is incorrect because it equates two pure numbers without any unit relationship.

C. 1 mm = 0.001 m
This is correct because the prefix "milli-" means 10^{-3}, so 1 mm = 10^{-3} m.

D. 1000 mm = 0.001 m
This is incorrect because 1000 mm = 1 m, not 0.001 m.

Q3. Which option lists an incorrect SI unit for the quantity?

A. Momentum: kg · m/s
This is correct because momentum has dimensions of mass times velocity, kg m/s.

B. Power: watt
This is correct because power is energy per time, measured in watts.

C. Acceleration: m^2/s^2
This is incorrect because acceleration is m/s^2, not m^2/s^2.

D. Pressure: N/m^2
This is correct because pressure is force per area, N/m^2 (pascal).

Q4. A scout is stationed at the origin of a map grid and spots an enemy camp at coordinates $\vec{r} = 4\hat{i} + 7\hat{j}$, where distances are measured in kilofurlongs. Which statement is correct?

A. The distance from the origin to the camp is 8 kilofurlongs.
This is incorrect because the true distance is $\sqrt{4^2 + 7^2} = \sqrt{65} \approx 8.06$, not exactly 8.

B. The distance from the origin to the camp is 7 kilofurlongs.
This is incorrect because 7 kilofurlongs is just the y-coordinate, not the radial distance.

C. The distance from the origin to the camp is 11 kilofurlongs.
This is incorrect because adding components $(4 + 7)$ is not the same as computing Euclidean distance.

D. None of the above is correct.
This is correct because the distance is $\sqrt{65} \approx 8.06$ kilofurlongs, which does not match 8, 7, or 11.

Q5. If the displacement of an object is given by $\vec{s} = \left((0.80\ \text{m/s}^3)t^3 + (3.6\ \text{m/s}^2)t^2\right)\hat{i}$, which statement is correct?

A. This motion has constant acceleration.
This is incorrect because differentiating twice gives an acceleration that depends on t.

B. The magnitude of the acceleration is 3.6 m/s^2.
This is incorrect because 3.6 m/s^2 is a coefficient in $s(t)$, not a constant value of $a(t)$.

C. The magnitude of the acceleration at $t = 0$ is 0 m/s^2.
This is incorrect because the constant term in $a(t)$ is nonzero.

D. The magnitude of the acceleration at $t = 0$ is 7.2 m/s^2.
This is correct because $a(t) = \frac{d^2s}{dt^2} = (4.8t + 7.2)\hat{i}$, so at $t = 0$ the magnitude is 7.2 m/s^2.

Q6. A robot moves in a straight line from position A $\left(\vec{r}_A = 1.0\hat{i} - 2.0\hat{j}\right)$ to position B $\left(\vec{r}_B = 7.0\hat{i} + 4.0\hat{j}\right)$ in 5 s, where length is in meters. Which statement is correct?

A. The distance traveled is 8.5 m.
This is correct because $|\vec{r}_B - \vec{r}_A| = \sqrt{6^2 + 6^2} = \sqrt{72} \approx 8.49\ \text{m} \approx 8.5\ \text{m}$.

B. The speed at $t = 0$ must be 1.7 m/s.
This is incorrect because the problem only fixes endpoints and travel time, not the instantaneous speed at $t = 0$.

C. The acceleration at $t = 0$ must be 0.68 m/s^2.
This is incorrect because no information about how velocity changes with time is given.

D. None of the above is correct.
This is incorrect because the straight-line distance statement in the first option is correct.

Q7. A ball is launched from level ground with initial speed v_0 at an angle θ above the horizontal (ignore air resistance). What is the ball's speed at its maximum height?

A. v_0
This is incorrect because the vertical component of velocity decreases to zero at the top, reducing the speed.

B. $v_0 \cos\theta$
This is correct because at maximum height $v_y = 0$ while the horizontal component remains $v_x = v_0 \cos\theta$.

C. $v_0 \sin\theta$
This is incorrect because $v_0 \sin\theta$ is the initial vertical component, not the remaining speed at the top.

D. 0
This is incorrect because only the vertical velocity becomes zero; the horizontal velocity generally remains nonzero.

Q8. A probe initially moves with velocity $\vec{v}_0 = (18.0\ \text{m/s})\hat{i}$. Its thrusters then provide a constant acceleration $\vec{a} = (1.5\ \text{m/s}^2)\hat{i} + (3.0\ \text{m/s}^2)\hat{j}$. Which statement is correct?

A. The x-component of the probe's velocity remains constant.
This is incorrect because a nonzero a_x causes v_x to change with time.

B. The y-component of the probe's velocity remains constant.
This is incorrect because a nonzero a_y causes v_y to change with time.

C. Both the direction of motion and the speed change over time.
This is correct because acceleration in both $\hat{i}$ and $\hat{j}$ changes the velocity components, altering both speed and direction.

D. None of the above is correct.
This is incorrect because the third option correctly describes the effect of a 2D acceleration.

Q9. Two point charges exert an electrostatic force on each other. If each charge's magnitude is multiplied by 4 and the separation distance is multiplied by 2, what happens to the magnitude of the force?

A. It stays the same.
This is incorrect because changing charges and separation changes $q_1 q_2 / r^2$ unless the factors cancel.

B. It decreases by a factor of 4.
This is incorrect because the charge increase dominates here rather than reducing the force.

C. It increases by a factor of 2.
This is incorrect because the net scaling is larger than 2 when both charges are quadrupled.

D. It increases by a factor of 4.
This is correct because $F \propto q_1 q_2 / r^2$, so $q_1 q_2 \to 16(q_1 q_2)$ and $r^2 \to 4r^2$, giving an overall factor of $16/4 = 4$.

E. It increases by a factor of 16.
This is incorrect because it ignores the reduction from the increased separation via the $1/r^2$ dependence.

Q10. The escape speed from a spherical planet is $v = \sqrt{\frac{2GM}{R}}$, where M is the planet's mass and R is its radius. Which statement is correct?

A. A more massive spacecraft needs a larger escape speed than a less massive spacecraft from the same planet.
This is incorrect because the formula depends on the planet's M and R, not on the spacecraft's mass.

B. Escape speed is higher on a planet with smaller mass and larger radius.
This is incorrect because smaller M and larger R both make $\sqrt{2GM/R}$ smaller.

C. Escape speed is lower on a planet with greater mass and smaller radius.
This is incorrect because greater M and smaller R both increase $\sqrt{2GM/R}$.

D. Escape speed is higher on a planet with greater mass and smaller radius.
This is correct because increasing M and decreasing R both increase $v = \sqrt{2GM/R}$.

Q11. A constant force $\vec{F} = (5.5\text{ N})\hat{i} + (-2.0\text{ N})\hat{j}$ acts on a particle as it moves along the x axis from $x = 1.2$ m to $x = 4.8$ m. What is the work done by the force on the particle?

A. 19.8 J
This is correct because the displacement is along $+x$ only, so $W = \vec{F} \cdot \Delta\vec{r} = F_x \Delta x = 5.5(4.8 - 1.2) = 19.8$ J.

B. 7.2 J
This is incorrect because it effectively uses only the y-component magnitude or an unrelated difference rather than $F_x \Delta x$.

C. 23.3 J
This is incorrect because it comes from using the force magnitude $|\vec{F}|$ instead of the dot product with the x-only displacement.

D. 0 J
This is incorrect because there is a nonzero x-component of force along the direction of motion.

Q12. A 3.2 kg cart moves toward a bumper at 4.0 m/s, hits it, and rebounds at 1.6 m/s in the opposite direction. The contact time is 0.25 s. What is the magnitude of the average force on the cart during the collision?

A. 71.7 N
This is correct because $|F_{\text{avg}}| = m|\Delta v|/\Delta t = 3.2\,|(-1.6 - 4.0)|/0.25 \approx 71.7$ N.

B. 28.7 N
This is incorrect because it effectively uses only one speed or forgets that reversing direction makes Δv larger.

C. 44.8 N
This is incorrect because it results from using $|\Delta v| = 4.0 - 1.6$ instead of $4.0 + 1.6$ when directions oppose.

D. None of the above is correct.
This is incorrect because the first option matches the impulse-based average-force calculation.

Q13. A 65 kg skateboarder is initially at rest. A 0.25 kg ball traveling at 30 m/s is caught and held (the ball and skateboarder move together afterward). What is the skateboarder's speed immediately after the catch?

A. 0.115 m/s
This is correct because momentum conservation for a perfectly inelastic catch gives $v = \frac{mv_0}{M+m} = \frac{0.25 \cdot 30}{65+0.25} \approx 0.115$ m/s.

B. 0.46 m/s
This is incorrect because it ignores the large mass of the skateboarder and effectively uses $v \approx v_0\, m/M$ with the wrong arithmetic.

C. 30 m/s
This is incorrect because it assumes the skateboarder simply takes on the ball's original speed, violating momentum conservation.

D. 7.5 m/s
This is incorrect because it comes from dividing momentum by the ball's mass instead of the combined mass.
